Seed Saving Field Manual

The complete reference for saving seeds from 8 vegetables

$14

Open-pollinated · Community-grown · Food sovereignty for everyone

Seed Saving Field Manual

A Practical Reference for the Independent Grower

Published by Seedwarden

This manual is built to be used in the field. Dog-ear it. Write in the margins. Come back to the tomato section every July.

Important Notice

This manual is educational content based on real seed saving experience. Your results will vary depending on your climate, growing conditions, and the specific varieties you work with. It is not professional agricultural advice. Use it as a practical reference, keep notes on what works in your situation, and expect some trial and error — that is part of the process.

Contents

1. Why Save Seeds

You already know why.

Cost. Seed packets are $3–6 each and climbing. A single well-saved tomato can yield 200+ seeds. One season of careful saving can eliminate your seed budget for years.

Adaptation. Seeds saved from your garden, in your soil, in your climate, gradually adapt to your conditions. After 3–5 generations, a variety is meaningfully different from the one in a catalog — better suited to your microclimate, your drainage, your season length. You can't buy that.

Variety preservation. Commercial seed catalogs stock what sells at scale. Thousands of regionally adapted varieties have already disappeared. The ones that still exist often live only in the hands of individual growers. Every seed library is a form of preservation.

Independence. Hybrid (F1) seeds are designed not to save reliably. That's not a conspiracy — it's a business model. Open-pollinated and heirloom varieties reproduce true from seed. Saving them is a straightforward act of growing your own inputs.

That's enough on the why. The rest of this manual is the how.

2. How Plants Reproduce

Understanding pollination is the foundation of seed saving. Get this right and everything else follows.

Self-Pollinating Plants

These plants pollinate themselves — pollen moves from the stamens to the pistil within the same flower, or between flowers on the same plant, before most insects have a chance to interfere. The result: seeds that reliably reproduce the parent plant.

Best for beginners. Cross-contamination risk is low. Isolation is minimal or unnecessary.

Vegetable	Notes
Tomatoes	Flowers self-pollinate before opening fu
Peppers	Almost entirely self-pollinating
Beans	Pollination occurs inside closed flower
Peas	Same as beans
Lettuce	Pollination happens before flower fully
Eggplant	Primarily self-pollinating

Cross-Pollinating Plants

These plants rely on insects, wind, or other external vectors to move pollen between plants — and often between varieties. If two varieties of the same species are growing nearby, their seeds may produce crosses.

Requires more care. You need either physical isolation (distance or barriers) or active management (hand pollination, caging) to keep varieties pure.

Vegetable	Pollinator	Cross Risk
Squash & Zucchini	Insects	Very high
Cucumbers	Insects	High
Corn	Wind	Very high
Brassicas (cabbage, kale, broccoli)	Insects	High
Sunflowers	Insects/wind	Moderate–high
Beets & Chard	Wind	High
Basil	Insects	Low–moderate

Why This Matters

A cross doesn't ruin the fruit you're eating this season. It affects the seeds. If your 'Black Beauty' zucchini cross-pollinated with your 'Yellow Crookneck' squash, the fruit may look normal — but the seeds inside will produce plants that are neither variety. The surprise usually shows up the following year.

The practical rule: **if you're saving seeds from a crop, know what's flowering nearby and whether cross-pollination is possible.**

What to Save From

Always save from **open-pollinated (OP)** or **heirloom** varieties — never F1 hybrids. Hybrid seeds are bred by crossing two distinct parent lines. Their offspring don't reproduce true — you'll get a wide range of unpredictable plants. Open-pollinated varieties, saved and grown over generations, will reliably reproduce their parent traits.

If you're not sure whether a variety is OP or hybrid: check the seed packet. F1 hybrids are always labeled as such. If a variety has a traditional name and no F1 designation, it's almost certainly open-pollinated.

3. Seed Saving by Vegetable — The Core Reference

This is the section to bookmark. Each entry covers everything you need for that crop: when to harvest, how to process, how to clean, isolation requirements, and any special considerations.

Tomatoes

Difficulty: Beginner
Pollination: Self-pollinating
Isolation needed: 10–25 feet between varieties (minimal risk; most cross-pollination in tomatoes is rare)

Tomatoes use wet processing — the seeds are surrounded by a fermentation-inhibiting gel that needs to be removed before storage. The fermentation method does this while also killing some seed-borne pathogens.

When to Harvest for Seed

Wait until the tomato is **fully ripe** — past eating ripeness if anything. For red varieties, you want deep, uniform color and slight softening. For varieties that don't change color much when ripe (some greens, some blacks), wait until they begin to soften at the blossom end. The seeds inside need to fully mature; harvesting early reduces viability.

Best practice: mark your best seed-saving candidates early in the season. Tie a piece of yarn or surveyor's tape around them. Leave them on the vine until they're slightly overripe. Don't eat them. Don't give them to anyone.

Extraction and Fermentation

1. Cut the tomato in half across the equator (not top to bottom). Squeeze or spoon the seed-containing gel into a small jar or cup. Add a roughly equal volume of water.
2. Cover loosely (a piece of paper towel held with a rubber band works well) and leave at room temperature, out of direct sunlight. 65–80°F is ideal.
3. Over 2–5 days, the mixture will ferment. You'll see mold growing on the surface — this is normal and expected. The fermentation breaks down the gel surrounding the seeds.
4. When the viable seeds have sunk to the bottom and the non-viable seeds and debris are floating, it's ready. Don't let it go longer than 5 days or seeds may begin to germinate.
5. Pour off the floating material, mold, and surface liquid. Add fresh water, swirl, and pour off again. Repeat until the water runs fairly clear and you're left with clean seeds at the bottom.
6. Pour onto a plate, a ceramic tile, or a piece of parchment paper. Spread in a single layer. Let dry for 1–2 weeks, stirring daily to prevent clumping.

Isolation Notes

Commercial growers maintain 35 feet or more between varieties, but in a home garden with normal plant spacing, 10–25 feet is generally sufficient. For practical purposes: if you grow only one tomato variety at a time, you need no isolation at all.

If you're growing multiple heirloom varieties and want to be precise, use a soft brush to hand-pollinate and mark those flowers, or simply save from plants spaced well apart from different-variety plants.

Special Notes

- Yield: One large tomato yields 30–100+ seeds. You rarely need more than 2–3 fruits for a full season's supply.
- Saving from multiple plants: For genetic diversity, save from at least 3–6 plants of the same variety.
- Selecting for traits: Choose fruits from plants that set early, produced heavily, showed no disease, and tasted best. You're selecting your next generation here.

Peppers

Difficulty: Beginner
Pollination: Self-pollinating (occasional insect cross-pollination possible)
Isolation needed: 150–300 feet for strict purity; 35 feet practical minimum in home gardens

Peppers are straightforward — dry processing only, no fermentation.

When to Harvest for Seed

Peppers change color as they mature — green to red, orange, yellow, purple, brown, depending on the variety. **Seeds are only fully mature in the final ripe color.** A green bell pepper contains immature seeds. Leave peppers on the plant until they reach their final ripe color and begin to soften.

If frost threatens before peppers are ripe, pull the whole plant and hang it in a warm, dry place. Peppers will continue to ripen off the vine.

Extraction and Drying

1. Cut the pepper open. The seeds are attached to the central core (placenta).
2. Use your fingers or a small knife to scrape seeds away from the core onto a dry plate or screen.
3. Hot pepper caution: wear gloves and avoid touching your face. Capsaicin doesn't wash off easily and will transfer to seeds.
4. Spread seeds in a single layer on a screen, plate, or paper towel. Keep in a warm, dry location with good airflow.
5. Dry for 1–2 weeks, turning occasionally. Seeds are ready when they snap cleanly rather than bending.

Isolation Notes

Peppers cross-pollinate less readily than squash or corn, but it does happen — primarily through bees. If you're growing sweet peppers and hot peppers in the same garden, there is a cross risk. However: the cross doesn't affect this year's fruit flavor. It only affects the seeds. If flavor purity in seeds matters to you, maintain at least 35 feet or cage flowering plants.

Special Notes

- Yield: One mature pepper contains 50–150+ seeds. You usually need only 1–2 peppers per variety.
- Sweet vs. hot crosses: Crossing sweet and hot peppers won't make this year's fruit spicy. Next year's plants from those seeds are unpredictable.
- Selecting: Choose seeds from the largest, earliest-ripening peppers on your most vigorous plants.

Beans and Peas

Difficulty: Beginner (easiest crop to save seeds from)
Pollination: Self-pollinating; flowers close before insects can interfere
Isolation needed: 10–25 feet (minimal risk in practice)

Beans and peas are among the most reliable crops for seed saving. The mechanics are simple: let them dry on the vine.

When to Harvest for Seed

Don't harvest these for eating — let the pods **dry completely on the plant.** You're looking for:

- Pods that have turned tan, brown, or papery
- Pods that rattle when shaken (seeds loose inside)
- Vines that have begun to die back

If wet weather or frost threatens before pods are fully dry, pull the entire plant and hang it upside down in a dry, well-ventilated space (a garage, shed, or covered porch). The plants will continue to dry and seeds will mature off the vine.

Extraction — Threshing

Small quantities: Simply shell by hand. Hold the pod over a bowl and split it open along the seam. The seeds fall out. Easy.

Larger quantities (threshing):

1. Place dried plants or pods in a pillowcase or cloth sack.
2. Beat the sack against a hard surface, or lay it on the ground and walk on it.
3. Open the sack — seeds and pod debris will be loose.
4. Winnow to separate (see Section 4).

Cleaning

Beans and peas typically need only minimal cleaning. Remove obviously broken or shriveled seeds. For bulk batches, a light winnowing pass removes most chaff.

Isolation Notes

Because beans and peas self-pollinate inside the closed flower, cross-pollination is rare but not impossible. For practical purposes: if you're growing only standard garden beans and peas, no special isolation is needed. If you're growing multiple varieties you want to keep strictly pure, maintain 25 feet of separation.

Runner beans (Phaseolus coccineus) are the exception — they are insect-pollinated and require 150+ feet between varieties for strict isolation.

Special Notes

- Seed is the same as the food. The dried bean or pea you'd cook is the seed. If you've ever stored dried beans in your pantry, you understand the basics of bean seed storage.
- Selecting: Save from the pods that set earliest, from the most vigorous plants, from beans that match your ideal size and color. Don't save from plants that showed disease.
- Filet beans and heirlooms: Some filet varieties can become starchy if left too long for seed. This is fine — the texture changes are for eating; the genetics are unaffected.

Lettuce

Difficulty: Beginner
Pollination: Self-pollinating (insect crosses possible but uncommon)
Isolation needed: 6–12 feet practical minimum; 12–25 feet for strict purity

Lettuce goes to seed (bolts) in response to heat and long days. The key challenge is timing the harvest right — the seeds ripen unevenly across the plant.

When to Harvest for Seed

Lettuce bolts when it sends up a tall flower stalk, typically 2–4 feet high. Small yellow flowers appear along the stalk and branches. After flowers fade, each becomes a small seed attached to a white feathery pappus (similar to a dandelion).

The seeds ripen from the bottom of the stalk upward over 1–3 weeks. This creates a window problem: by the time upper seeds are ripe, lower seeds may already be dropping.

Signs of ripeness:

- The white pappus is visible and fluffy
- Seeds come loose easily when you run a finger along the stem
- The stalk is beginning to dry

Harvest Methods

Daily harvest (best quality): Check plants every 1–2 days once the first seeds begin dropping. Run your fingers up the stalks and collect ripe seeds into a paper bag. This captures seeds at peak ripeness before they drop.

Bag method: When seeds begin to ripen, tie a paper bag loosely over the top of the stalk. As seeds mature and fall, they collect in the bag. Check every few days and empty into your processing container.

Full stalk harvest: Cut the entire stalk when roughly 50–60% of seeds appear ripe. Lay in a paper bag in a dry place for a week — remaining seeds will continue to mature and drop.

Cleaning

Place harvested material in a bowl and crumble lightly to separate seeds from stems. The feathery pappus and chaff can be removed by light winnowing. Seeds are small and brown/tan when ripe.

Isolation Notes

Lettuce crosses are possible but not common in home gardens. Different varieties of lettuce in the same garden rarely produce true crosses in practice. For strict purity, 6 feet between varieties is usually adequate. Different lettuce types (loose-leaf vs. romaine vs. butterhead) will still cross if they bolt simultaneously.

Special Notes

- Bitter bolted leaves: Once lettuce bolts, the leaves become bitter and inedible. This is expected — the plant is putting energy into seeds, not leaves. This doesn't affect seed quality.
- Timing: To save seed from spring-planted lettuce, you may need to leave plants in the ground through summer heat. In hot climates, plan a specific seed-saving plant rather than trying to save from every plant.
- Yield: One well-developed plant produces hundreds of seeds. You rarely need more than 3–5 plants to have a full season's supply.

Squash and Zucchini

Difficulty: Intermediate

Pollination: Insect-pollinated (bees required)

Isolation needed: 500–1,500 feet between varieties of the same species; hand pollination is the practical alternative

Squash is where most beginners run into trouble. The cross-pollination risk is real, and the species boundaries matter.

Understanding Squash Species

Not all squash cross with each other. Cross-pollination only occurs within the same species. The main species:

Species	Common Varieties
Cucurbita pepo	Zucchini, acorn, delicata, spaghetti, pu
Cucurbita maxima	Buttercup, Hubbard, kabocha, Atlantic Gi
Cucurbita moschata	Butternut, Long Island Cheese pumpkin
Cucurbita argyrosperma	Cushaw squash

Critical: A zucchini and a butternut cannot cross — they're different species. But a zucchini and an acorn squash can, because both are *C. pepo*.

If you're growing only one variety per species, you have no cross risk.

When to Harvest for Seed

Leave the squash on the vine far longer than you would for eating. Seeds need additional time to mature after the flesh is edible.

- Summer squash (zucchini): Let the fruit grow to full mature size — typically 12–18 inches and with a hardening skin. This is a "seed squash," not a table squash. Leave it for at least 4–6 weeks after it would normally be harvested for food.
- Winter squash: Harvest after the vine dies back naturally, or at least 8 weeks after fruit set. Cure for 2–4 weeks after harvest before extracting seeds.

Seed Extraction

1. Cut the squash open. Scoop out the seed cavity.
2. Place seeds and pulp in a bowl of water. Separate seeds from pulp by hand — viable seeds sink, pulp floats.
3. Rinse seeds thoroughly.
4. Spread on a screen or non-stick surface (not paper towels — they stick). Dry in a single layer for 2–4 weeks.

Hand Pollination (for Purity Without Isolation)

If you can't maintain 500+ feet between varieties of the same species, hand pollination is your best option.

1. The evening before: Identify female flowers (have a small proto-fruit at the base) and male flowers that will open the next morning. Both should be about to open — puffy and showing color at the tip.
2. Tape or rubber-band the flowers closed the evening before they open. This prevents insects from getting in overnight.
3. The next morning: Open the male flower. Use a small paintbrush or your finger to collect pollen, then open the female flower and apply pollen directly to the stigma (the central structure). Re-tape the female flower closed after pollinating.
4. Mark the pollinated fruit with tape or a tag. This is your seed squash.
5. Remove the tape from the female flower after 24 hours. The fruit should begin developing normally.

Isolation Notes

If relying on distance rather than hand pollination: 500 feet is the practical minimum for most gardens; 1,500 feet for strict purity. In a typical urban or suburban setting with neighbors growing squash, hand pollination is more realistic than distance.

Special Notes

- This year's fruit is not affected by crosses. Crosses only show up in seeds and the following year's plants.
- One fruit per plant for seed: You only need one mature seed squash per variety. Once you've designated it, you can harvest everything else for eating normally.

Cucumbers

Difficulty: Intermediate
Pollination: Insect-pollinated
Isolation needed: 500–1,000 feet; or hand pollination (same technique as squash)

Cucumbers are in their own genus (Cucumis sativus) and only cross with other cucumbers — not with squash or melons. If you're growing only one cucumber variety, you have no cross risk.

When to Harvest for Seed

Do not harvest seed cucumbers for eating. Leave them on the vine until they are **fully mature** — well past the eating stage. A ripe seed cucumber:

- Changes color from green to yellow, orange, or white (depending on variety)
- Becomes soft and slightly mushy
- Has seeds that are hard and fully formed inside

This typically requires leaving the cucumber on the vine for 5–6 weeks after it would normally be picked for the table.

Extraction

1. Cut the cucumber open lengthwise.
2. Scoop seeds and gel into a bowl.
3. Add water and use your fingers to separate seeds from gel.
4. Viable seeds sink; non-viable seeds and debris float. Pour off the floaters.
5. Rinse, drain, and spread seeds to dry on a screen or non-stick surface. Dry 2–4 weeks.

Fermentation (Optional but Recommended)

Cucumbers can be processed the same way as tomatoes — the fermentation method removes the gel coating more thoroughly and kills some seed-borne diseases. Add equal water to seeds and gel, ferment 1–3 days, skim and rinse clean. Cucumber fermentation moves faster than tomato; don't go beyond 3 days.

Isolation Notes

If growing multiple cucumber varieties, hand pollination is the practical solution (same technique as squash: tape flowers closed the night before, hand-pollinate in the morning). 500 feet of distance is the other option.

Special Notes

- One fruit per plant is enough. A single mature cucumber contains 200–500 seeds.
- Parthenocarpic varieties (seedless or near-seedless cucumbers) are not suitable for seed saving.
- Cucumber vs. other Cucumis: Cucumbers will not cross with cantaloupe or muskmelon (C. melo) — different species.

Basil

Difficulty: Beginner
Pollination: Insect-pollinated; some self-pollination
Isolation needed: 50–150 feet between varieties; less critical than squash

Basil is straightforward — let it flower, wait for seeds to mature, harvest before seeds drop.

When to Harvest for Seed

Basil produces small flower spikes (called racemes) covered in tiny flowers. As flowers fade, small seed pods form at the base of each flower. Seeds ripen from the bottom of the spike upward.

Signs of seed ripeness:

- Flowers have browned and dropped
- Seed capsules at the base of the spike are brown and papery
- Seeds inside are dark (brown to black)

If you wait until all seeds are ripe, the first-ripe seeds will have already dropped. The practical approach: harvest spikes when roughly 60–70% of the seeds appear ripe and mature.

Harvest and Extraction

1. Cut seed spikes from the plant.
2. Let them dry in a paper bag in a warm, dry place for 1–2 weeks.
3. Once fully dry, crumble the spikes between your hands over a bowl. Seeds will fall out.
4. Use light winnowing or a fine-mesh screen to separate seeds from chaff.

Basil seeds are very small — dark, oval, about 1–2mm. They'll look like small black sesame seeds.

Growing Notes for Seed Saving

If you want to harvest basil leaves all season and save seeds:

- Designate specific plants for seed saving early. Don't pinch their flowers.
- Continue pinching all other plants for harvest.
- Seed-saving plants will become woody and stop producing leaves worth eating. This is expected.

Isolation Notes

Different basil varieties (Genovese, Thai, Purple, Lemon, etc.) can cross. If you're growing multiple varieties and care about purity, maintain 50–150 feet between them or isolate with a physical barrier. However, for most home growers keeping 1–2 varieties, this isn't a practical issue.

Sunflowers

Difficulty: Beginner to Intermediate

Pollination: Insect-pollinated (and some wind)

Isolation needed: 1,500 feet for strict purity; 300–500 feet practical minimum; or isolation with caging

When to Harvest for Seed

Sunflower seeds mature inside the head after the petals have dropped. Maturity signs:

- The back of the head turns from green to yellow to brown
- Seeds in the head are plump and have developed color (black, striped, etc.)
- The head begins to droop under seed weight
- The disc florets (tiny flowers in the center) dry from the outside in — when the center is dry, seeds are mature

Harvest timing matters: harvesting too early means immature seeds. Too late and you lose seeds to birds or the head begins to rot in wet weather.

Bird and Wildlife Protection

Birds find sunflower heads before you do. Options:

- Paper bag method: Once petals drop and seeds begin filling, loosely tie a paper bag (not plastic — causes rot) over the head. This blocks birds while allowing airflow.
- Netting: Cut a piece of fine mesh bird netting and tie it around the head.
- Harvest early and dry indoors: Cut the head with 6–12 inches of stem when the back is yellow-brown. Hang or lay flat in a warm, dry, ventilated space. Seeds will continue to mature.

Drying the Head

After harvest, hang heads or lay them on a screen in a warm, dry location with good airflow. Let them dry for 2–4 weeks. The head will shrink and contract, loosening seeds.

Extracting Seeds

Once the head is fully dry:

- Rub two heads together over a bucket — seeds knock loose easily
- Or rub your palm firmly across the face of the head
- For large heads, a stiff brush works well

Cleaning

The chaff (small dried flower parts) mixes with seeds. Winnow by pouring seeds from one bucket to another in front of a fan or in a light breeze. Chaff blows away; seeds fall.

Isolation Notes

For strict variety purity, sunflowers need serious distance — 1,500 feet or more, because they're visited heavily by bees. In a typical garden setting, this is impractical. If purity matters, use a floating row cover or fine mesh fabric to cage flowering heads before they open, then hand-pollinate (self-pollinate within the same variety by moving pollen from plant to plant with a brush, within the cage).

For most home growers maintaining a single sunflower variety, isolation isn't needed.

Special Notes

- Edible vs. ornamental: Standard large sunflowers (Helianthus annuus) save easily. Many ornamental multi-headed varieties are open-pollinated and can be saved. Check that your variety is not F1 hybrid.
- Perennial sunflowers (H. tuberosus, Jerusalem artichoke) reproduce vegetatively from tubers — seeds aren't the primary propagation method.

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4. Cleaning and Processing Seeds

All seed saving involves some level of cleaning — removing chaff, non-viable seeds, and debris before drying and storage.

Wet vs. Dry Processing

Wet processing is used for seeds enclosed in a moist, fleshy fruit: tomatoes, cucumbers, squash, peppers (though peppers are usually processed dry). The wet process uses water to separate seeds from flesh and, in some cases, fermentation to remove seed coatings.

Dry processing is used for seeds that mature in dry pods or husks: beans, peas, lettuce, sunflowers, basil, corn. These seeds air-dry on the plant and are extracted by threshing and winnowing.

The Float Test

Water is a quick way to assess viability in seeds large enough to float test. Fill a bowl with water and add your cleaned seeds.

- Seeds that sink: Dense, fully developed, likely viable
- Seeds that float: Hollow, underdeveloped, or non-viable — discard

Limitations: The float test works well for tomatoes, squash, cucumbers, and beans. It is less reliable for very small seeds (lettuce, basil) and doesn't work at all for seeds that have been fully dried (dried seeds float regardless of viability).

Use the float test on freshly cleaned wet seeds before the drying phase.

Winnowing

Winnowing separates seeds from chaff using air movement. The basic principle: seeds are heavier than chaff and fall faster in a moving airstream.

Simple winnowing method:

1. Hold a wide bowl at about waist height.
2. Pour seeds slowly from a second container held above your head.
3. A light breeze (or a fan on low setting) carries the chaff away while seeds fall into the bowl.
4. Repeat 2–3 times for cleaner results.

Adjust technique: Move containers closer together for heavier seeds (beans, sunflowers); farther apart for lighter seeds (lettuce, basil) so there's more time for wind to carry the chaff.

Screen method: Stack two screens — one with larger holes (lets seeds fall through, catches large chaff) and one with smaller holes (catches seeds, lets fine dust through). Run seeds through once or twice. This works especially well for beans and corn.

Removing Debris by Hand

For small batches, handpicking is often the fastest method. Spread cleaned seeds on a white plate or light surface — discolored, shriveled, cracked, or otherwise damaged seeds are easy to spot and remove.

5. Drying Seeds Properly

Moisture is the primary enemy of stored seed. Seeds that go into storage with excess moisture will mold, lose viability, and may heat up and die entirely. Proper drying is non-negotiable.

Why It Matters

Seeds are alive, just sleeping. Keep them cool and dry and they stay that way. At 50% relative humidity, seeds contain about 12–13% moisture — adequate for short-term storage. For long-term storage, you want 6–8% moisture content, achievable only with active drying.

A seed stored at high humidity can lose half its viability in a single season.

Air Drying

The baseline method. Works well in most climates if done correctly.

Requirements:

- Warm temperature (65–80°F)
- Low humidity (under 60% RH)
- Good airflow — not sealed in a container
- Indirect light (UV degrades seeds over time)

Process:

1. After wet-processing, spread seeds in a single layer on a non-stick surface: ceramic plates, glass, screens, or parchment paper. Avoid paper towels — small seeds stick permanently.
2. Stir or turn seeds daily to prevent clumping and ensure even drying.
3. Duration depends on seed size: small seeds (lettuce, basil) — 1 week; medium seeds (tomatoes, peppers) — 1–2 weeks; large seeds (beans, squash) — 2–4 weeks.

Test for dryness: Seeds should snap or shatter when bent or bitten, not flex. A tomato seed that bends is not dry enough. A tomato seed that snaps and breaks is ready.

The Silica Gel Method

For storage in humid climates, or for seeds you want to keep for 5+ years, silica gel is the most reliable way to get seeds to storage-ready dryness.

What you need:

- Indicating silica gel (blue or orange crystals that change color when saturated; orange → green, blue → pink)
- A sealed container (glass jar with tight lid, Tupperware)

Process:

1. Air-dry seeds first for at least 1 week.
2. Place air-dried seeds in paper envelopes (not sealed plastic).
3. Put envelopes and a layer of silica gel in the container in a 1:10 ratio by weight (1 part silica gel to 10 parts seeds is a guideline; for small batches, a tablespoon of silica gel per jar is sufficient).
4. Seal and leave for 1–2 weeks.
5. Check the color indicator. If the gel has changed color (saturated), replace with fresh dry gel.

Recharging silica gel: Spread saturated gel on a baking sheet in a 250°F oven for 1–2 hours. It returns to its original color and is ready to reuse.

How Dry Is Dry Enough?

Seed Type	Air Dry Time (70°F, 50% RH)	Silica Gel Finish
Tomato	10–14 days	Optional but recommended
Pepper	10–14 days	Optional
Bean / Pea	2–4 weeks on vine, then 1–2 weeks indoors	Usually not needed
Lettuce	7–10 days	Recommended (small seeds absorb moisture
Squash / Pumpkin	2–4 weeks	Recommended for long storage
Sunflower	2–4 weeks (head)	Optional

The snap test is your best field check: a properly dried seed snaps, shatters, or crumbles when bent firmly. A seed with too much moisture will flex.

What Not to Do

- Don't use a dehydrator on high heat. Temperatures above 95°F can kill seeds. If you use a dehydrator, use the lowest setting (95°F or below) and monitor closely.
- Don't dry in plastic bags. Plastic traps moisture. Use screens, plates, or paper.
- Don't rush to store. An extra week of drying time costs nothing; mold in storage costs everything.

6. Storage: Making Seeds Last

A well-organized seed library is one of the most useful tools a grower can have. Getting the storage conditions right makes the difference between seeds that last 2 years and seeds that last 10.

The Three Enemies of Stored Seed

1. Moisture — the primary threat. Triggers premature germination, mold, and enzymatic degradation.
2. Heat — accelerates aging. Seeds stored at 40°F last roughly twice as long as seeds stored at 70°F.
3. Light — UV degrades seed viability over time. Store in opaque or dark containers, or in a dark location.

Container Options

Paper envelopes — Best for active use and short-term storage (1–3 years). Breathable, cheap, easy to label, easy to open repeatedly. Vulnerable to moisture and pests.

Glass jars with tight lids — Excellent for medium and long-term storage. Airtight, moisture-resistant, easy to see contents. Add a small silica gel packet inside for extra protection. Mason jars work perfectly.

Mylar bags with heat seal — Best for very long-term storage (5–10+ years). Nearly impermeable to moisture and oxygen. Requires a heat sealer. Overkill for most home growers, but worth considering for irreplaceable varieties or bulk storage.

What not to use: Zip-lock bags are acceptable for short-term use but are not truly airtight and allow moisture exchange over time. Avoid storing seeds in plastic bags long-term.

Labeling: What to Include

A seed that isn't labeled is close to useless. Label every packet at the time you process it, before you put it away.

Minimum required information:

Field	Why It Matters
Variety name	Obvious
Crop	Especially important when seeds look sim
Year harvested	Viability degrades; you need to know how
Source	Whether you saved them or traded; the so

Optional but valuable:

- Number of seeds in packet
- Isolation method used (hand-pollinated, caged, distance)
- Notes on the parent plant (yield, disease resistance, flavor, days to maturity)
- Germination test result and date

Write labels in permanent marker. If using envelopes inside a jar, label both the envelope and the jar lid.

Temperature and Humidity Guidelines

Storage Condition	Expected Viability
70°F / 50% RH (room temperature)	1–3 years (most seeds)
50°F / 40% RH (cool basement or root cel	3–5 years
40°F / 35% RH (dedicated refrigerator)	5–10 years
32°F / <30% RH (freezer, properly dried)	10–20+ years

For freezer storage: Seeds must be at 6–8% moisture content before freezing. Wet seeds will be killed by ice crystal formation. Seal completely before placing in the freezer, and allow sealed containers to come to room temperature before opening — this prevents condensation from forming on the cold seeds.

A cool basement or spare refrigerator dedicated to seeds (set to 40–45°F) is practical for most home growers and provides excellent longevity without the handling complexity of freezer storage.

Seed Viability by Vegetable

Use this table to prioritize what to test and replant. Viability indicates years of storage under good conditions before germination rate drops significantly.

Vegetable	Average Viability	Notes
Onion / Leek	1–2 years	Short-lived; save fresh every year
Parsnip	1–2 years	Buy or save fresh annually
Sweet Corn	2–3 years	Declines fast; test regularly
Pepper	2–4 years	Dry well; viable longer with silica gel
Lettuce	3–5 years	
Basil	3–5 years	
Tomato	4–6 years	
Bean / Pea	3–5 years	
Squash / Pumpkin	4–6 years	
Cucumber	5–7 years	
Sunflower	3–5 years	
Brassicas (kale, cabbage)	4–5 years	
Beet / Chard	4–6 years	
Carrot	3–4 years	
Celery / Celeriac	3–5 years	
Watermelon	4–6 years	

These are guidelines, not guarantees. How you store them matters more than the numbers. Always test germination rates before the season.

7. Testing Germination Before Planting

Don't find out in May that your lettuce seeds have 10% germination. Test in February.

The Paper Towel Test

Materials: Paper towels, a zip-lock bag or plate, warm water, a marker.

Process:

1. Dampen a paper towel — thoroughly wet, then wrung out so it's not dripping.
2. Lay 10 seeds of a single variety in a row on one half of the towel.
3. Fold the other half over the seeds.
4. Place in a zip-lock bag or on a plate covered with plastic wrap.
5. Label with variety name and date.
6. Store at 65–75°F. Avoid direct sunlight.
7. Check every 24–48 hours. Keep the towel moist throughout.

Timing: Most vegetables germinate within 5–14 days under good conditions. Wait for the full expected germination window before counting.

Vegetable	Days to Check
Lettuce	3–7 days
Tomato	5–10 days
Pepper	7–14 days (can be slow)
Bean / Pea	3–7 days
Squash / Cucumber	3–7 days
Basil	4–8 days
Sunflower	3–7 days

Counting: After the full window, count how many of your 10 seeds germinated. This number is your germination percentage.

- 10/10 = 100% — excellent
- 8/10 = 80% — good; plant at normal spacing
- 5–7/10 = 50–70% — acceptable; sow more densely
- Below 5/10 = under 50% — replace or sow very heavily; this lot is declining

When to Replace Your Stock

A germination rate below 50% is the trigger to either:

1. Sow very densely and thin — works for crops where seeds are cheap and abundant
2. Save fresh seed this season — the priority if it's a variety you care about
3. Source replacement seed — if you don't have a current crop growing

Important: Don't discard old seeds entirely until you've saved fresh replacements. Even 30–40% germination can produce a crop if you sow heavily. Low-viability seeds are far better than no seeds in an emergency.

Germination Rate Doesn't Tell You Everything

A seed can germinate and still produce a weak plant. When assessing old stock, also note:

- Vigor — do germinated seeds grow strongly and quickly, or weakly and slowly?
- Uniformity — are all seedlings similar, or highly variable?

Weak or non-uniform germination from old stock often indicates declining seed health even when raw germination rate is acceptable.

8. Building Your Seed Library Over Time

The difference between a seed collector and a seed grower is selection. Saving seeds passively — grabbing whatever fruit is at hand — produces seeds that will grow. Saving seeds intentionally, season over season, produces seeds that get better.

Organizing Your Collection

Physical organization:

Keep seeds in a single dedicated container — a box, a small crate, or a file box. Organize by crop family or alphabetically, whichever you'll actually use. Loose seeds in a drawer or scattered across a shelf are seeds you won't be able to find.

A shoebox with labeled dividers (TOMATOES / PEPPERS / BRASSICAS / ALLIUMS / BEANS / etc.) works as well as any commercial system.

Inventory:

Once a year — before the growing season, when you're planning — go through your collection. Note what you have, what year it's from, and the approximate quantity. This takes 20–30 minutes and prevents planting failures from forgotten old stock.

A simple spreadsheet or a handwritten list on the inside of the box lid does the job:

```
TOMATOES
  Cherokee Purple — 2024 — ~80 seeds — good germ (8/10 Feb 2025)
  Brandywine — 2023 — ~40 seeds — test before planting
  Green Zebra — 2025 — ~120 seeds — fresh, untested
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Selecting the Best Specimens

Every time you save seed, you're making a selection decision. The seeds you save determine what your next generation looks like.

What to select for:

- Earliest to mature — saves the crop in short seasons; especially important in regions with early frost
- Disease resistance — save from plants that stayed clean when others didn't
- Yield — save from the most productive plants
- Flavor and appearance — for eating crops, the obvious metric
- Adaptability — plants that handled your specific stress conditions (drought, heat, wet feet) without failing

Practical process:

1. Early in the season, mark your best 3–6 plants of each variety with a stake or piece of tape.
2. These marked plants get their best fruit reserved for seed. Everything else on those plants can still be eaten, but the designated seed fruits stay on the vine until fully ripe.
3. At harvest, note on your label what you selected for ("saved from earliest and largest fruits, no blight").

Over 5–7 generations of consistent selection, you can noticeably shift a variety's performance in your conditions. This is landrace development at the home-garden scale — the same process that produced most traditional heirloom varieties to begin with.

Maintaining Genetic Diversity

Saving from a single plant, season after season, narrows the genetics of your strain. In a small population, bad traits can get locked in by chance.

Minimum plant numbers for seed saving:

Crop	Minimum Plants to Save From
Self-pollinating (tomato, pepper, bean)	3–6 plants
Insect-pollinated (squash, cucumber)	6–12 plants
Wind-pollinated (corn, beet)	50–200 plants

For corn especially, saving from fewer than 50 plants produces significant inbreeding depression within a few generations. If you're growing a corn variety for seed, grow a full block and save from as many plants as possible.

For tomatoes and beans, 3–6 plants is genuinely sufficient. These crops have been maintained in small family gardens for generations.

Trading and Swapping

A seed library is more resilient when connected to other growers. Trading seeds:

- Brings new genetics into your collection
- Gives you backup copies of your own varieties held by others
- Spreads regionally-adapted varieties through your community

Seed swaps: Local gardening clubs, community gardens, and agricultural extension offices often host seed swaps in late winter or early spring. These are among the best places to find locally-adapted varieties that won't be in any catalog.

Online exchanges: Seed Savers Exchange, Reddit's r/seedswap, and various regional seed libraries facilitate trades by mail. When sending seeds by mail, use small paper envelopes inside a padded envelope with a rigid card to prevent crushing.

What to bring to a swap:

- Clearly labeled packets (variety, crop, year saved, isolation method if relevant)
- More than you think you need — it's better to give generously
- Any interesting or unusual varieties you've maintained

What to ask when receiving seeds:

- Is this open-pollinated?
- How old are these seeds (what year were they saved)?
- Have they been isolation-grown, or is there possible cross-pollination?

Year-Over-Year Records

The growers who maintain the most reliable seed stocks are the ones who keep notes. You don't need elaborate systems. A small notebook dedicated to the garden — or a few pages at the back of a garden journal — is enough.

What's worth recording:

- What you planted and from what seed lot
- Notable events: disease pressure, drought, early frost, outstanding yield
- What you saved from and why (best plants, best fruits)
- Germination test results
- Any notable changes in the variety compared to previous years

After 5–10 years of records, patterns become visible: which varieties perform consistently for you, which require constant replanting from outside sources, which have actually improved under your selection. That record is more valuable than any catalog.

Quick Reference: Isolation Distances

Crop	Minimum (Home Garden)	Strict Purity
Tomato	10–25 ft	35 ft
Pepper	35 ft	150–300 ft
Bean (bush/pole)	10–25 ft	25 ft
Runner Bean	150 ft	300 ft
Pea	10 ft	25 ft
Lettuce	6–12 ft	25 ft
Squash (same species)	Hand pollinate	500–1,500 ft
Cucumber	Hand pollinate	500–1,000 ft
Basil	50 ft	150 ft
Sunflower	300–500 ft	1,500 ft
Corn	Not practical	1,500–2,000 ft
Brassicas	Not practical	1,500 ft
Beet / Chard	1 mile	—

Isolation distances assume open, relatively flat terrain. Hedges, buildings, and other obstacles reduce effective cross-pollination risk.

Quick Reference: Seed Processing Methods

Crop	Processing	Fermentation?	Typical Dry Time
Tomato	Wet	Yes (2–5 days)	10–14 days
Pepper	Dry	No	10–14 days
Bean / Pea	Dry (on vine)	No	Finish indoors 1–2 weeks
Lettuce	Dry	No	7–10 days
Squash	Wet	Optional	2–4 weeks
Cucumber	Wet	Optional (1–3 days)	2–4 weeks
Basil	Dry	No	7–10 days after stalk harvest
Sunflower	Dry (head)	No	2–4 weeks (head drying)

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More from Seedwarden

If you found this guide useful, you might also like:

- Food Sovereignty Starter Guide ($8) — The big-picture roadmap for growing, saving, and sharing your own food
- Heirloom Variety Selection Guide ($11) — Pick the right open-pollinated varieties for your specific soil, climate, and light
- Seed Swap Hosting Kit ($10) — Everything you need to turn saved seeds into a neighborhood tradition

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